# Limitations of Automated Binary Vulnerability Analysis in Optimized and Stripped Binaries

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Milestone Report

## 1 Revised scope and thesis

The proposal asked whether automated static analysis and LLM-based vulnerability detectors stay reliable once binaries are compiled with aggressive optimization and stripped of symbols. Three changes were made after instructor feedback and a closer read of the literature.

First, the baseline changed. The proposal named `cppcheck`, but `cppcheck` analyzes C source and never inspects a compiled binary, so it cannot measure any effect of optimization or stripping. It is kept only as a source-level reference point; the binary-level static baseline is now `cwe_checker`, which operates on the lifted binary and covers the three target weakness classes. Second, the targets are reproduction-grade: each program has a vulnerable and a fixed (patched) version, so false positives and hallucination can be measured, not only missed bugs. Third, optimization and symbol stripping are now varied *independently* in a $2 \times 2$ design, as the feedback requested, rather than bundled into a single `-O3 -s` build.

**Thesis (current).** We reproduce a published source-level vulnerability-detection result and test how far it degrades when the same programs are presented as decompiled binaries, isolating the effect of compiler optimization (`-O0` vs. `-O3`) from the effect of symbol stripping (kept vs. stripped), in order to mark the point at which automated triage is no longer reliable and manual Ghidra/GDB analysis becomes mandatory.

## 2 Reproduction target and baseline

We reproduce a specific result from Khare et al. [1], who evaluated 16 LLMs on five vulnerability datasets. We adopt their *Juliet C/C++* setting and their exact prompts (system prompt, Basic, and CWE-specific) at temperature 0. The claim we reproduce is their reported Juliet C/C++ detection performance, summarized in Table 1. Their best model on this dataset (Qwen-2.5-14B, CWE-specific prompt) reaches accuracy 0.65 and F1 0.72, while the CodeQL static baseline reaches accuracy 0.72 but recall only 0.44. A notable pattern is that most models achieve high recall ($\sim$0.9–1.0) with precision near 0.5: they call almost everything vulnerable. Our fixed-version programs are designed to expose exactly this bias and to track whether it worsens on binaries.

| Source (Juliet C/C++) | Method | Acc | Prec | Rec | F1 |
|---|---|---|---|---|---|
| Khare et al. [1] | CodeQL (static) | 0.72 | 0.99 | 0.44 | 0.61 |
| Khare et al. [1] | Qwen-2.5-14B (CWE) | 0.65 | 0.60 | 0.89 | 0.72 |
| Khare et al. [1] | GPT-4 (CWE) | 0.58 | 0.54 | 0.95 | 0.69 |
| This work (reproduction) | **[model]** source | **[acc]** | **[prec]** | **[rec]** | **[f1]** |

Table 1: Published Juliet C/C++ numbers we reproduce, and our source-level reproduction row (to be filled from `results/summary.json`).

## 3 What is implemented

The full pipeline runs end to end; only the model inference step requires the local GPU and is run on the author's machine (Ollama, **[model]**, temperature 0). The stages are:

- **Targets.** Eight C programs, each with a vulnerable and a fixed variant (16 source cases), covering buffer overflow (CWE-121/787), format string (CWE-134), and use-after-free / double-free (CWE-416/415). Juliet-style identifiers are renamed to neutral names so the model cannot read the answer from a symbol like `CWE121_..._bad`.
- **Build matrix.** `compile.sh` produces the $2 \times 2$ builds per variant. Stripping is applied to a *copy* of the compiled binary (`strip --strip-all`), not via recompilation, so the machine code is byte-identical to its symbol-bearing twin. `_FORTIFY_SOURCE` and the stack protector are disabled so `-O3` does not replace `strcpy` with `__strcpy_chk` or insert canaries, which would change the bug rather than merely optimize around it.
- **Build verification.** `verify_builds.sh` confirms (1) each stripped binary has a `.text` section byte-identical to its kept twin, so symbols are the only difference, and (2) every vulnerable binary still contains its bug after compilation — memory bugs confirmed by AddressSanitizer, format strings confirmed by a crash under a format-specifier payload — at both `-O0` and `-O3`. All 16 vulnerable builds are confirmed.
- **Decompilation.** `decompile.sh` exports Ghidra headless pseudo-C per binary (`objdump -d .text` fallback when Ghidra is absent), plus a comment-stripped source corpus for the reproduction condition.
- **Inference + scoring.** `run_pipeline.py` applies the Khare prompts and logs a structured verdict per item; `score.py` computes accuracy, precision, recall, F1, CWE-match, and hallucination per condition, and the factor-isolation deltas.

## 4 Evaluation design and the development/evaluation split

The detection task is binary (vulnerable vs. not), scored against the known label of each variant. Measures: accuracy, precision, recall, and F1 for detection; *CWE-match* (of correctly flagged vulnerable cases, the fraction assigned the right CWE); and *hallucination* (fraction of fixed programs flagged vulnerable). The five conditions are `source`, and the four binary builds `O0/kept`, `O0/stripped`, `O3/kept`, `O3/stripped`. Because the project reproduces and diagnoses rather than trains, the "split" is a prompt-development vs. evaluation split: prompts and parsing were fixed on two held-out programs (one overflow, one use-after-free) and never changed based on the remaining evaluation targets. Factor isolation compares conditions that differ in exactly one variable: `O0/kept` vs. `O3/kept` isolates optimization; `O0/kept` vs. `O0/stripped` isolates stripping.

## 5 Preliminary results and interpretation

*The pipeline, baseline reproduction, and verification are complete; the numbers below are filled from `results/summary.json` after the model run.* Table 2 reports detection and triage metrics per condition.

| Condition | Acc | Prec | Rec | F1 | CWE-match | Halluc. |
|---|---|---|---|---|---|---|
| source | [] | [] | [] | [] | [] | [] |
| O0 / kept | [] | [] | [] | [] | [] | [] |
| O0 / stripped | [] | [] | [] | [] | [] | [] |
| O3 / kept | [] | [] | [] | [] | [] | [] |
| O3 / stripped | [] | [] | [] | [] | [] | [] |

Table 2: Detection and triage metrics per condition (16 cases each).

**Expected reading (hypothesis, to confirm).** We expect F1 and CWE-match to be highest on `source`, to fall most under stripping (loss of function and variable names), and to fall further

under `-O3` (inlining and restructuring). The factor-isolation deltas will say which of the two matters more and whether they compound: if the `O0/kept→O3/stripped` drop exceeds the sum of the two single-factor drops, the factors interact, which would itself be a reportable finding. A *negative* result — little change across conditions — would contradict the thesis and would be reported as such.

## 6 What will be studied next (the distribution shift)

The stress condition is the structural distribution shift from source to `-O3`-stripped binary: identical semantics, radically different surface form. The remaining work is: (1) run `cwe_checker` on all four binary builds as the non-LLM binary baseline; (2) define the triage decision rule concretely — an alert is trusted only above a recall threshold with LLM and `cwe_checker` agreement, else it is routed to manual Ghidra/GDB — and report how often each condition would be routed to manual analysis; (3) a short qualitative failure analysis mapping specific compiler transformations (inlining, `strcpy` vectorization, dead-store elimination) to specific triage errors.

## 7 Completion plan

- **By Fri (milestone):** pipeline + baseline reproduction + verification done (complete); source and binary runs logged; Table 2 filled.
- **Final report:** add `cwe_checker` baseline; triage decision rule and routing rates; failure analysis; expand targets if time allows.
- **Risks:** a bug optimized away at `-O3` (mitigated by the ASan survival check, all 16 currently pass); Ghidra headless setup (mitigated by the `objdump` fallback, already working).

## Contributions

Solo project. Los Escalante: target curation, build and verification scripts, decompilation pipeline, prompt pipeline and scorer, reproduction of the Khare et al. baseline, and this report.

## Artifacts

Code and data: **[repo or drive link]**. Scripts: `compile.sh`, `verify_builds.sh`, `decompile.sh`, `run_pipeline.py`, `score.py`.

## References


[1] A. Khare, S. Dutta, Z. Li, A. Solko-Breslin, R. Alur, M. Naik. "Understanding the Effectiveness of Large Language Models in Detecting Security Vulnerabilities." arXiv:2311.16169, 2024 (ICST 2025).

[2] T. Boland, P. E. Black. "Juliet 1.1 C/C++ and Java Test Suite." *IEEE Computer*, 2012. NIST SAMATE SARD.

[3] Fraunhofer FKIE. "cwe_checker." `https://github.com/fkie-cad/cwe_checker`.